

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Summer 2024

Course: B. Tech. Branch : Electrical Engineering and Allied Semester :VI

Subject Code & Name: BTEEPE604B Smart Grid Technology

Max Marks: 60

Date:21/06/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		
A) Write a key challenges and policies related to smart grids in India.	K1/CO1	6
B) Describe the Intelligent Electronic Devices (IED) with their application.	K2/CO1	6
C) Illustrate the Pumped Hydro used for renewable energy smart storage.	K3/CO1	6
Q.2 Solve Any Two of the following.		
A) Describe Architecture of Smart Grid with suitable layout diagram.	K2/CO2	6
B) Explain the following terms: A) Transmission Automation in a Smart Grid B) Distribution Automation in a Smart Grid	K2/CO2	6
C) Explain how artificial intelligence techniques are being utilized in smart grid development.	K2/CO2	6
Q.3 Solve Any Two of the following.		
A) Discuss the Environmental impact and Climate Change due to electrical power generation.	K2/CO3	6
B) Explain in the details Electric Vehicles and plug-in hybrids contribution in micro grids.	K2/CO3	6
C) Illustrate working of Microgrid with its benefits and drawbacks.	K3/CO3	6
Q.4 Solve Any Two of the following.		
A) Explain the role of IP-based systems and Power Line Communications (PLC) in smart grid communication networks.	K2/CO4	6
B) Explain the role of Advanced Metering Infrastructure (AMI) in communication technology.	K2/CO4	6
C) Illustrate working of Phasor Measurement Units (PMUs) with suitable diagram.	K3/CO4	6
Q.5 Solve Any Two of the following.		
A) Explain the Load Frequency Control (LFC) and Voltage Control in Micro Grid System.	K2/CO5	6
B) Discuss various Cyber Security Challenges in Smart Grid.	K2/CO5	6
C) Illustrate how the test beds are used for different case studies of smart grids.	K3/CO5	6

***** End *****